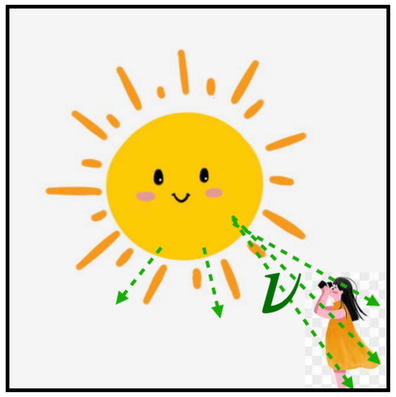
Ghostly Yardsticks: Understanding Nuclei with Neutrinos

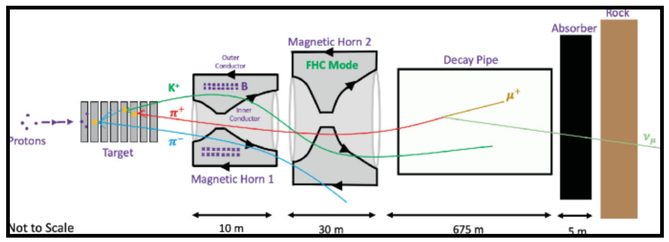
Nimmy Sarah Alex | University of Rochester | MINERvA Collaboration

Neutrinos: The Most Elusive Ghosts



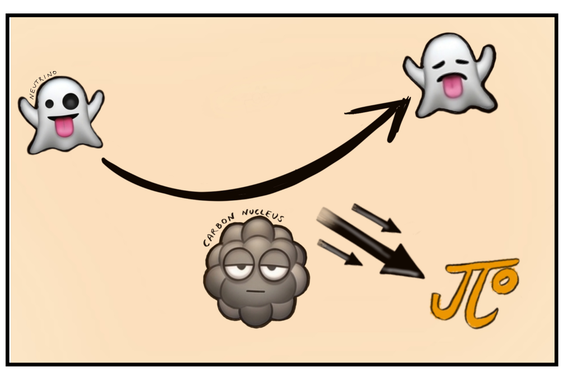
100 billion neutrinos pass through your thumb every second

How do we study their properties?



We produce them using accelerators and study different interactions

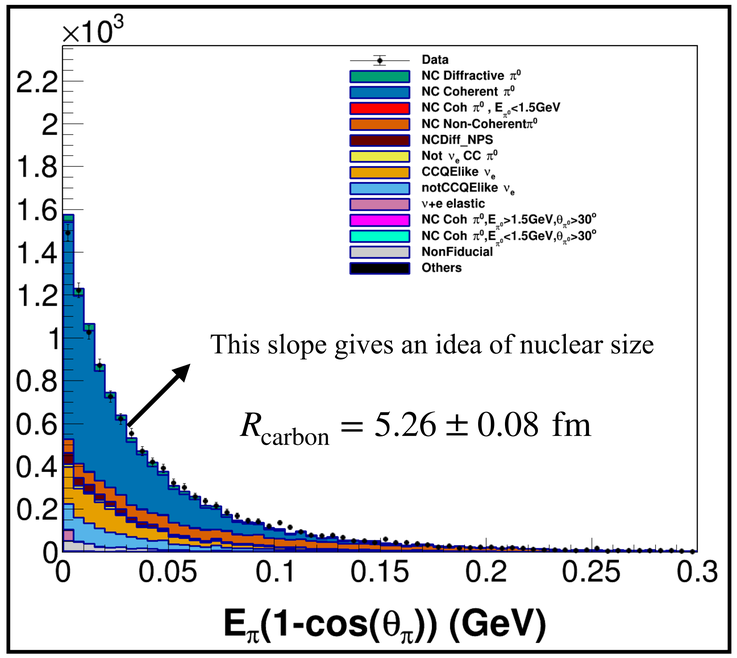
What is the interaction of interest?

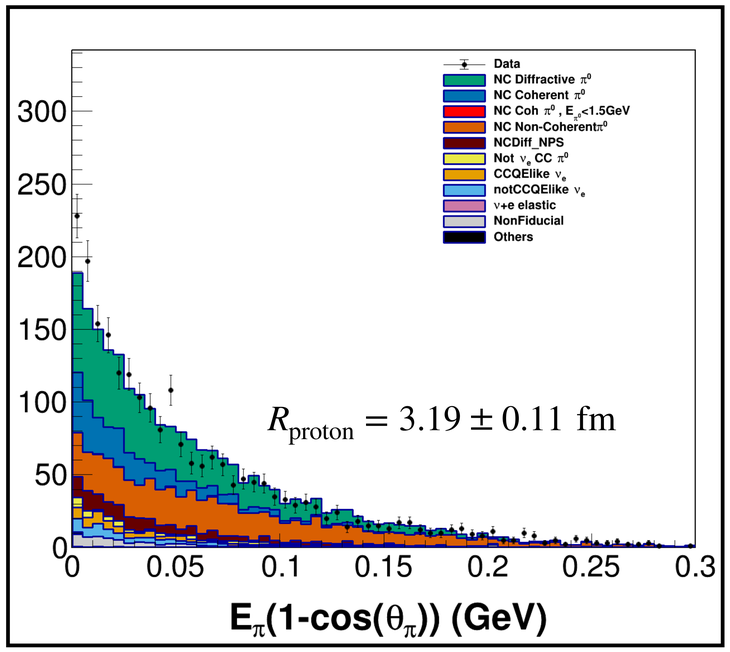


Neutral Current Pion Production Process
Neutrino interacts with a carbon or hydrogen nuclei producing a neutral pion and leaving nucleus intact in its ground state

Measurements on Neutral Current Coherent π^0 production processes assumes Coherence Condition

Measure of momentum transferred → $E_{\pi^0}(1 - cos\theta_{\pi^0}) \leq \hbar c/R$ ← The resolution





Previous studies of neutral current coherent π^0 production assumes the coherence condition.

These studies did not distinguish interactions on hydrogen from other multi-nucleon nuclei.

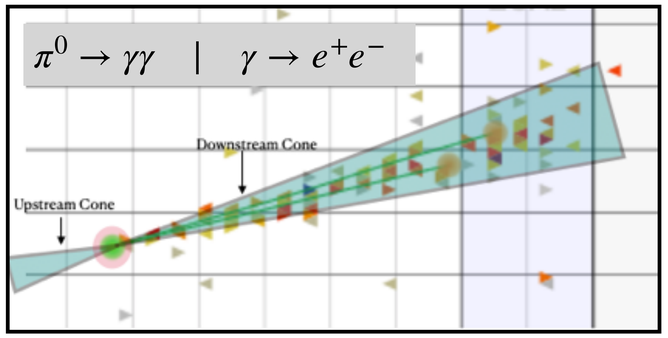
This analysis tests the coherence condition on both carbon and hydrogen to verify these long-standing assumptions.

Why is this interaction interesting?

Low momentum transfer ↔ higher wavelength

Wavelength comparable in size to the nucleus can probe the entire nucleus
Analogy: A larger wave engulfs a whole boat as whole, whereas a smaller ripple probes specific parts

How do we actually study them?



MINERvA identifies π^0 signals and looks at the momentum distribution for these specific events

Further Reading and References